ZIM School - Use Cases

07. Nutrition: baseline and endline weighings and the SF8

Modules: Nutrition
3 use cases (changes) · 4 views and reports
Written from the use case commands, handlers, policies and rules in `src/` on 2026-11-10.

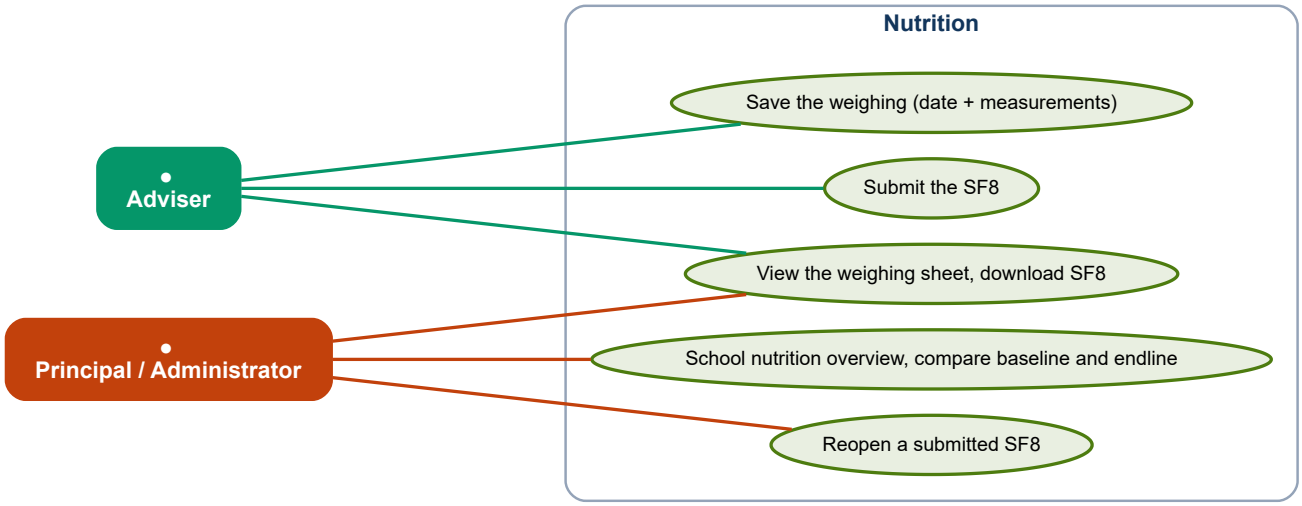
1. What this area does

Twice a school year (**baseline** and **endline**) the adviser weighs the class: the date of weighing plus each learner's weight (kg) and height (m), or a remark such as "absent". From these the SF8 report (School Form 8, Learner's Basic Health and Nutrition Report) computes each learner's BMI and classifies **BMI-for-age** and **height-for-age** with the WHO growth tables (`resources/who`); the classification is computed when the report is built, not stored.

Submitting the SF8 hands it to the principal and locks the period; the principal or an administrator can reopen it. Who counts is the section's roster on the date of weighing. Every change takes the `nutrition-section-<id>` lock. Who may do what is `NutritionPolicy`: only the adviser fills it (current year only); the principal and administrators view it and reopen it. The SF8 prints the school's SF8 certifier from the school profile.

2. Use case map

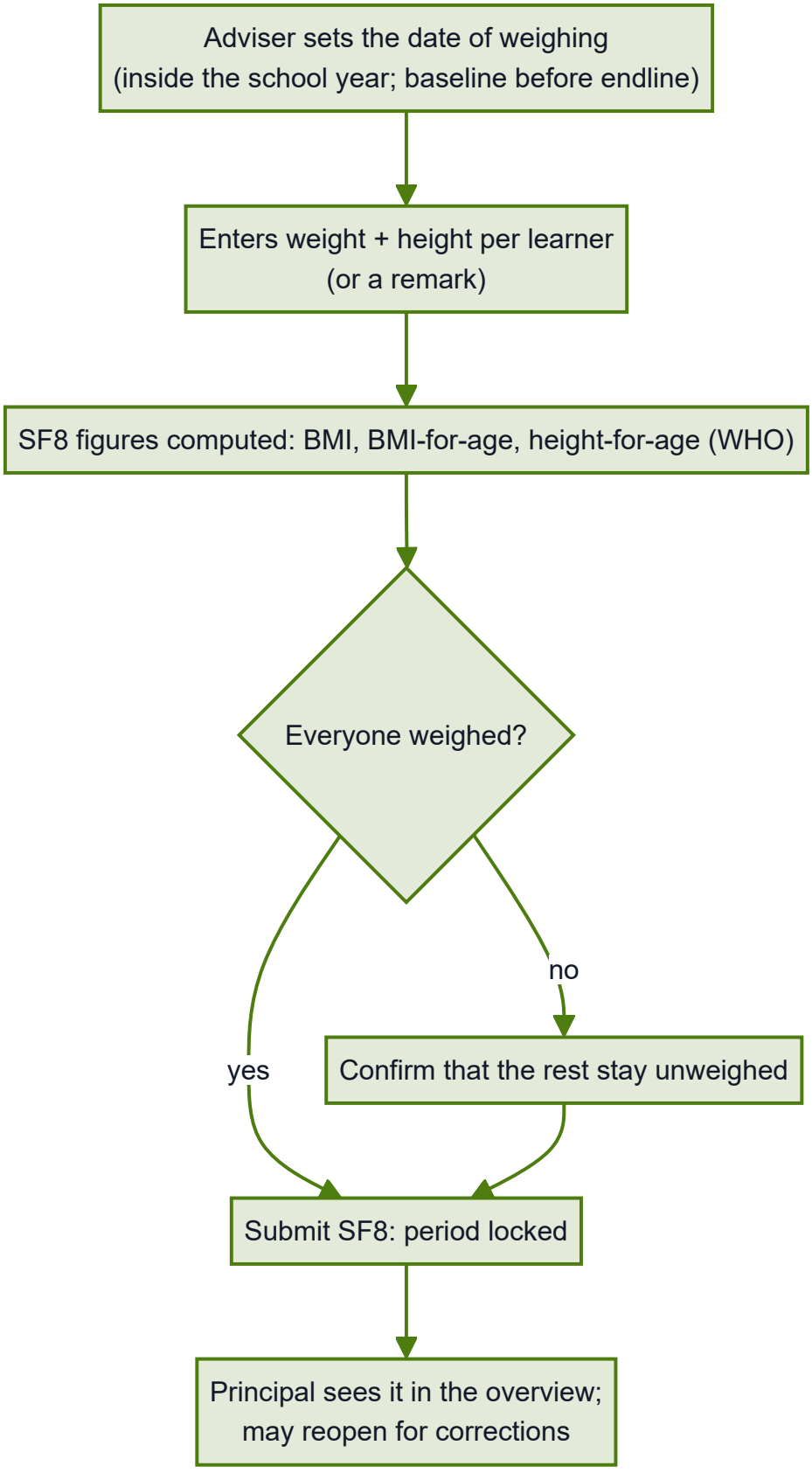
A standard use case diagram: the actors on the left, each use case as an ellipse inside the system frame, and a line for each use case an actor starts. A dashed line has a note on when it applies.



Adviser · Principal / Administrator — Line colour = the actor who starts it.

Nutrition — Ellipse colour = the module the use case belongs to.

3.1 Flow: A weighing period



Nutrition Drawn in the colour of the module this flow belongs to.

4. Use cases (changes)

Each card is one use case command sent through the use case bus. It runs in one database transaction: if any rule refuses, nothing is saved and the message in *Alternatives and refusals* is shown to the user.

NUT-01 Save the weighing

Primary actor	Adviser
Goal	Record the date of weighing and each learner's weight and height for the baseline or endline.
Where (route)	`teacher_class_nutrition_period /teacher/class/nutrition/{period}`
Preconditions	• The teacher advises a section this school year; the period is not submitted.
Main flow	1. The adviser opens the period, sets the date of weighing and fills the measurements (remarks may stand alone). 2. Saves. The system takes the section lock, checks the date and stores the assessment and its measurements. 3. Learners who don't count on the date of weighing are ignored.
Alternatives and refusals	• "Weigh inside the school year (<range>)." • "The endline must be after the baseline (<date>)." / "The baseline must be before the endline (<date>)." • "<period> is submitted and locked." • "You can no longer weigh this section." • Busy: "Another change is being saved. Please try again."
Result (data changed)	NutritionAssessment (section, period, weighed_on) and NutritionMeasurement rows.
In the code	`SaveWeighingCommand (MeasurementInput) → NutritionRecorder::save`

NUT-02 Submit the SF8

Primary actor	Adviser
Goal	Hand the period's SF8 to the principal.
Where (route)	`teacher_class_nutrition_submit /teacher/class/nutrition/{period}/submit`
Preconditions	• The weighing is saved; not yet submitted.
Main flow	1. The adviser presses Submit; if some learners have no weight and height, they confirm. 2. The period is locked and "Submitted SF8 of ..." is logged.
Alternatives and refusals	• "Set the date of weighing and save first." • "<period> is already submitted." • "N learners have no weight and height. Confirm to submit anyway."
Result (data changed)	Sf8Submission (section, period).
In the code	`SubmitSf8Command`

NUT-03 Reopen a submitted SF8

Primary actor	Principal, Administrator
Goal	Let the adviser correct a submitted period.
Where (route)	`school_nutrition_reopen` `/schools/{id}/sections/{section}/nutrition/{period}/reopen`
Preconditions	• Manages the school; the period is submitted.
Main flow	1. The principal presses Reopen on the section's sheet; "Reopened SF8 ..." is logged.
Alternatives and refusals	• "<period> is not submitted."
Result (data changed)	Sf8Submission removed.
In the code	`ReopenSf8Command`

5. Views, downloads and reports (read only)

Each row is one query sent through the query bus. Queries change nothing.

Page / file	Who	Route	What it shows	Query
Nutrition index	Adviser	`teacher_class_nutrition` `/teacher/class/nutrition`	The section with both periods' state and summary.	`GetNutritionIndexCommand`
Weighing sheet	Adviser, Principal	`teacher_class_nutrition_period; school_nutrition_section`	A period's measurements and classifications, and what the user may do.	`GetNutritionPeriodCommand`
School nutrition overview	Principal, Administrator	`school_nutrition` `/schools/{id}/nutrition` `(principal_nutrition)`	Every section's SF8 for a period, or baseline vs endline side by side.	`GetNutritionOverviewCommand`
SF8 file (.xlsx)	Adviser, Principal	`teacher_class_sf8 and the school's SF8 download`	DepEd SF8 filled into sf8-template.xlsx.	`GetSf8FileCommand`